Disentangled Foreground-Semantic Adapter Network for Generalized Aerial Image Few-Shot Semantic Segmentation

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***Abstract*— Semantic segmentation of remote sensing imagery requires extensive annotated samples for training, facing challenges in adapting to novel classes with few annotations. Few-shot semantic segmentation (FS-Seg) employs a support-to-query paradigm, which encounters many practical constraints. Recently, generalized few-shot semantic segmentation (GFS-Seg) has been proposed to align with general semantic segmentation paradigms, enabling the segmentation of all classes (both base and novel classes) in the image. However, existing GFS-Seg methods struggle with a large intra-class variance of background, degradation on base classes, and overfitting on novel classes during fine-tuning in aerial imagery. To address the above issues, we propose the disentangled foreground-semantic adapter network (DFSA-Net) for generalized aerial image FS-Seg. Specifically, to reduce the interference from background features, DFSA-Net employs a foreground-semantic decoder (FSD) to decompose semantic segmentation into foreground aggregation and multiclass refinement. To mitigate the base classes degradation and novel class overfitting during fine-tuning, we propose disentangled low-rank adapter (DLA) for fine-tuning phase, designed to preserve the base parameters while ensuring efficient adaptation to novel classes. Finally, we introduce an inference ensemble strategy that merges base and novel decoder prediction to achieve final output. Experimental results on NWPU and iSAID datasets demonstrate the superiority of our DFSA-Net over other compared methods.**

***Index Terms*— Adapter tuning, aerial imagery, deep learning, disentanglement representation, few-shot semantic segmentation (FS-Seg).**

I. Introduction

WITH the advancement of remote sensing technologies, remote sensing images achieve higher spatial resolution with detailed ground features, facilitating pixel-wise landcover classification, i.e., semantic segmentation. As shown in the first line of Fig. 1, semantic segmentation is a fundamental task for automatic scene understanding in remote sensing images, with applications from land resources statistics to Martian terrain exploration [1], [2], [3]. Despite the considerable progress achieved in semantic segmentation, existing methods heavily rely on extensive pixel-level annotations, which are both time-consuming and labor-intensive to obtain. To address this challenge, the few-shot semantic segmentation (denoted as FS-Seg) demonstrates great potential for various applications.

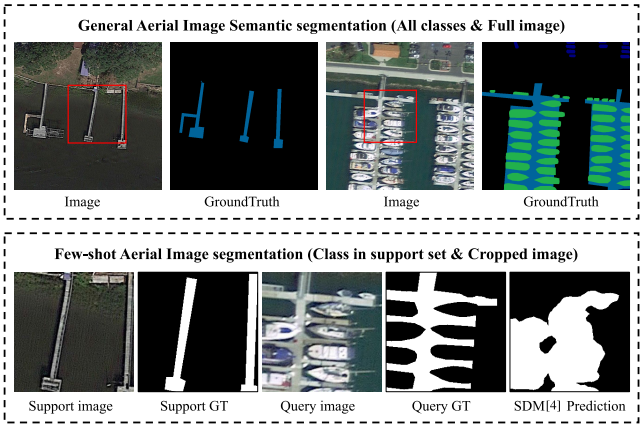


Fig. 1. Illustration of the difference between general and few-shot aerial image semantic segmentation. General aerial image semantic segmentation benchmark takes a full image as input and outputs mask of all classes in training data. In contrast, few-shot aerial image segmentation [4] adopts the support-to-query paradigm, which processes cropped images with foreground objects and is restricted to segment classes present in the support set.

Current FS-Seg methods primarily treat segmentation as a task of foreground detection within a support-to-query paradigm, organizing data into support and query sets for processing. As shown in Fig. 2(a), FS-Seg employs a dual-branch structure, processing triplets inputs (support images, support masks, and query images) to generate the query target mask defined in the support set. Typically, CANet [5] employs a dual-branch dense comparison module and an iterative optimization process for multilevel feature comparison and result refinement. For aerial image few-shot semantic segmentation, SDM [4] introduces a detailed matching module to achieve precise pixel-level matching between query and support feature, aiming to mitigate intra-class variation in aerial imagery. HMRE [6] proposes mutual representation enhancement between two branches to mitigate the intra-class variances and background interference problem. While existing methods achieve satisfactory performance by enhancing the matching capabilities between query and support features,

Manuscript received 19 April 2024; revised 4 August 2024 and 25 August 2024; accepted 2 September 2024. Date of publication 23 September 2024; date of current version 2 October 2024. This work was supported by the National Science and Technology Major Project under Grant 2022ZD0117400. *(Corresponding author: Jihao Yin.)*

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Digital Object Identifier 10.1109/TGRS.2024.3455427

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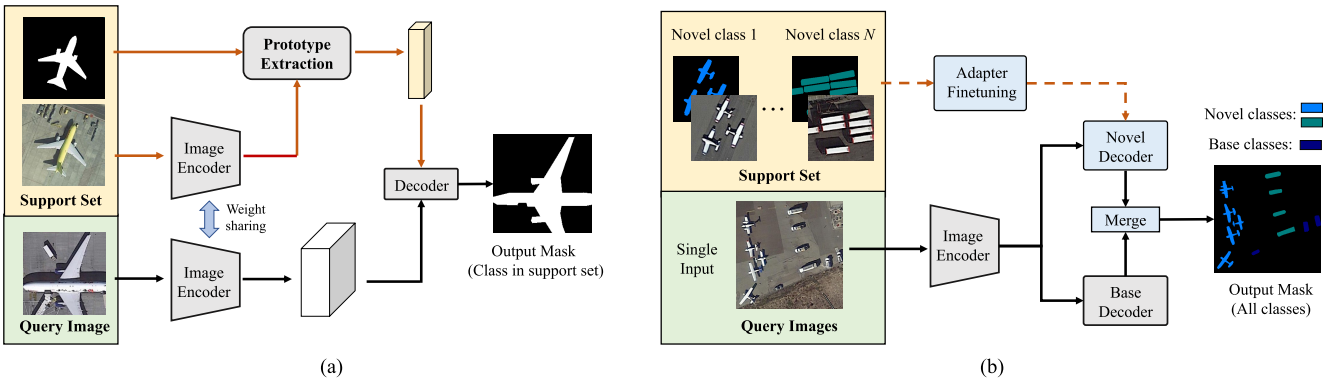


Fig. 2. Overview of few-shot segmentation paradigm for aerial imagery (a) illustrates the previous aerial image FS-Seg benchmark focused on segmenting specific classes defined within the support set, and (b) depicts the aerial image GFS-Seg benchmark, which extends capabilities to segment both base and novel classes without requiring paired support images during inference. The dotted line denotes that support sets are used only in the training process. (a) Previous aerial image FS-Seg benchmark and (b) generalized aerial image FS-Seg benchmark.

current FS-Seg methods for aerial imagery encounter two practical challenges.

1) These methods are restricted to segmenting only the novel classes within the support set, requiring that both support and query samples contain these matching novel classes. This dependence on specific prior knowledge is often impractical. Additionally, these methods do not consider the inclusion of base classes in query samples. For instance, as depicted in Fig. 1, the query ground-truth features only the harbor in the image, disregarding the base training category of ships.
2) Prevalent benchmarks for aerial image FS-Seg utilize cropped images (256×256) containing foreground objects, emphasizing class-agnostic foreground segmentation. Such emphasis neglects essential contextual landcover cues, which results in overfitting to training foreground object. It can be observed in Fig. 1 that the SDM [4] prediction inaccurately classifies ships as part of the harbor, demonstrating a misunderstanding of the harbor's query mask.

To bridge the gap between FS-Seg and traditional semantic segmentation, the concept of generalized few-shot semantic segmentation (GFS-Seg) [7], [8] has been introduced. GFS-Seg facilitates the segmentation of all classes (both base and novel) in an image without the need for paired support images during inference. In general, the GFS-Seg approach contains two distinct training phases: base class pretraining and novel class fine-tuning. Initially, the models are trained with comprehensive annotations from base classes to accurately segment pixels of these categories. In the subsequent phase, the models are fine-tuned using a limited set of labeled examples from novel classes, enabling the identification of pixels belonging to new categories, as illustrated in Fig. 2(b). During inference, GFS-Seg adopts the standard semantic segmentation paradigm, processing an image as input and producing a segmentation mask for all classes. Although these GFS-Seg methods demonstrate promising outcomes on natural images, applying them to aerial imagery for the GFS-Seg task faces unique obstacles. The challenges arise from two main factors: first, the significant intra-class variance in backgrounds impedes discriminative learning for foreground object categories; second, there is a notable decline in performance on base classes, alongside an increase in overfitting to novel classes during fine-tuning.

To alleviate the above two issues for generalized aerial image FS-Seg, we propose the disentangled foreground-semantic adapter network (DFSA-Net). Specifically, to reduce the interference from background features, DFSA-Net leverages a foreground-semantic decoder (FSD). This decoder effectively splits semantic segmentation into tasks of foreground aggregation and multiclass semantic refinement, thereby enhancing the ability to distinguish foreground categories under few-shot conditions. Moreover, to tackle the dual issues of base class degradation and novel class overfitting during fine-tuning, we propose a disentangled low-rank adapter (DLA) for the fine-tuning phase. Notably, to improve learning for novel classes, we implement salience suppression (SS) on base classes foreground in the fine-tuning phase, ensuring that the foreground is solely composed of novel class elements during the fine-tuning phase. The proposed DLA ensures the preservation of the pretrained model's integrity while offering a targeted solution to overfitting through low-rank parameter adjustments for novel class adaptation. Finally, we introduce an ensemble strategy that merges base and novel decoder prediction to achieve the final output.

To summarize, our main contributions are as follows.

1) To unify aerial image FS-Seg with general semantic segmentation, we propose DFSA-Net for aerial image GFS-Seg, significantly enhancing the utility of few-shot segmentation for aerial imagery.
2) We propose an FSD designed to suppress background feature interference. By decomposing semantic segmentation into foreground aggregation and multiclass semantic refinement, FSD effectively concentrates the learning on essential foreground category discrimination.
3) To address the challenges of base class degradation and novel class overfitting during the fine-tuning phase, we introduce the DLA. This strategy not only preserves the integrity of the pretrained model but also promotes effective adaptation by adopting low-rank parameters during novel class fine-tuning.
4) We incorporate FSD, DLA, and an ensemble strategy within DFSA-Net. Comprehensive experiments on

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NWPU and iSAID datasets validate the efficacy of our proposed DFSA-Net. Both quantitative and qualitative assessments indicate our model surpasses existing state-of-the-art FS-Seg methods.

The remainder of this article is structured as follows. Related works are briefly reviewed in Section II. Section III introduces the proposed DFSA-Net for aerial image GFS-Seg in detail. The effectiveness of the proposed method and ablation studies are demonstrated in Section IV by experimental results on two real high-resolution aerial imagery datasets. Finally, Section V concludes this article and suggests future research directions.

II. Related Works

A. Aerial Image Semantic Segmentation

Semantic segmentation is a fundamental task that aims to predict each pixel's class in an image. In recent years, the breakthrough of deep learning has led to eminent progress in semantic segmentation. Specifically, the fully convolutional network (FCN) was the first model designed for semantic segmentation, with many improved versions [9], [10], [11]. Early efforts focused on keeping the details of spatial information [12], [13], [14], [15]. Typically, U-Net [12] and feature pyramid network (FPN) [13] merges features at various levels to keep details while adding context. Another category of improvements aimed at enhancing the models' receptive fields [10], [11], [16], [17], [18], [19], [20], [21], [22]. To exploit multiscale contextual information, PPM [16] generates multiscale pyramidal feature maps via pyramid pooling, and deeplabv3+ [10] adopts atrous convolutions with different dilated rate to extract features with different receptive field. To achieve global context, Segmenter [11] jointly processes the patch and class tokens during the decoding phase, which generates dynamic prototype classifier for pixel segmentation. HRViT [23] enhances VIT to learn semantically rich and spatially precise multiscale representations by integrating high-resolution multibranch architectures.

Compared to natural images, aerial images are taken from a bird's eye view, featuring unique structural patterns that result in significant scale differences and complex background intra-class variations. To address the challenges of scale variation and scene diversity in aerial imagery, AERFC [24] adaptively controls convolution sampling locations and adjusts effective receptive fields to reduce training difficulty and preserve image details. ScasNet [25] introduces a self-cascaded encoder–decoder network that enhances segmentation through a sequential approach, progressively aggregating context from global to local scales and incorporating object refinement sub-networks. Meanwhile, DDCM-Net [26] leverages dilated convolutions with varied dilation rates, effectively expanding the network's receptive field to capture broader contextual information. RAFCN [20] introduces both spatial and channel relation modules to understand the relationships between any two positions within an image. MPCNet [27] introduces a multiscale prototype contrastive loss to foster more discriminative feature learning across different scales. Although these approaches yield high segmentation accuracy, they are constrained by the need for extensive annotated datasets. Additionally, they struggle to generalize to unseen target objects. To address these challenges, we unify FS-Seg with general semantic segmentation to improve the generalization capabilities.

B. Few-Shot Semantic Segmentation

FS-Seg task seeks to identify new semantic objects in query images with the aid of a few annotated support images. Typically, PANet [28] introduces a prototype alignment regularization to derive more generalized prototypes, significantly boosting parsing accuracy. CANet [5] employs a dual-branch dense comparison module and an iterative optimization process for multilevel feature comparison and result refinement. PMMs [29] introduce a parameter-free approach to create multiple prototypes, using varied prototypes to enhance segmentation performance through more comprehensive feature representations. Subsequently, PFENet [30] introduces a training-free prior mask, markedly enhancing the model's ability to generalize to unseen categories. HDMNet [31] presents a unique matching module based on correlation mechanisms and distillation strategies, aiming to mitigate the issue of overfitting to training classes while boosting the model's overall generalization capabilities. As for aerial imagery FS-Seg, SDM [4] introduces a detailed matching module to facilitate precise pixel-level matching between query and support, aiming to mitigate intra-class variation in aerial imagery. HMRE [6] constructs mutual representation enhancement between two branches to mitigate the intra-class variances and background interference problem. However, current FS-Seg methods are limited to segmenting novel classes in the support set, necessitating matching classes in support and query samples—a requirement that is frequently impractical. Additionally, current aerial image FS-Seg benchmarks focus on cropped images with foreground objects, leading to class-agnostic foreground segmentation. This approach causes overfitting to base classes and overlooks crucial contextual landcover cues.

Recently, the concept of GFS-Seg [7] has been introduced, which aims to segment both base and novel classes directly from query images in a single inference step, bypassing the need for paired support images. To achieve this goal, CAPL [7] utilizes contextual cues extracted from both support and query samples to enhance the classifier's ability to segment novel classes. Fine-tuning [32] has emerged as another viable and efficient strategy for tailoring models to recognize novel classes. POP [8] introduces the innovative technique of Projection onto Orthogonal Prototypes, proposing a methodical approach to extend the model's generalization to novel classes while minimally impacting its segmentation performance on base classes within the GFS-Seg framework. However, applying these methods to aerial imagery introduces specific challenges: considerable intra-class background variance impedes discriminative learning for foreground categories, and there is a noticeable decline in base class performance along with heightened overfitting to novel classes in the fine-tuning stage.

C. Attention Mechanism

Attention has been identified as a significant factor in visual perception, playing a crucial role in semantic segmentation [33], [34], [35]. Self-attention mechanisms [34], [35]

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and their variants [18], [36], [37], [38], [39], [40], [41] are integrated into decoders to aggregate global relational context information. OCNet [19], for example, generates a weighted attention map for each pixel, enhancing feature representation by amalgamating information from all positions. CBAM [36] sequentially infers attention maps along channel and spatial dimensions to improve network capacity. To mitigate the computational demand of the self-attention module, The asymmetric non-local neural network [39] utilizes a set of global descriptors for reconstructing feature maps, moving beyond the conventional approach of using individual pixels as reconstruction descriptors. Overall, as a fundamental operation in the widely researched Transformer architecture, the attention mechanism demonstrates excellent feature-capturing abilities for modeling long-range dependencies across various domains [42].

Notably, foreground attention is particularly emphasized in aerial image segmentation. FarSeg [43] improves foreground modeling and feature representation for occluded objects through a foreground-aware relation network, reducing false alarms in high-resolution images. FactSeg [44] introduces a foreground activation-driven framework for segmenting small objects, aiming to enhance segmentation accuracy for such targets. Additionally, HFGNet [45] underlines the importance of background context in boosting foreground saliency features to improve segmentation results. CCENet [46] designed a cascade class enhancement module to exploit class-aware prior information from the previous stage, thereby enhancing foreground salience. Moreover, MCCANet [47] introduces a multiscale channel-wise cross-attention network designed to eliminate semantic gaps in the presence of complex backgrounds and severe intra-class differences. Our proposed FSD is inspired by above developments, but we adopt a different feature aggregation module by additional loss constrain on intermediate foreground prediction.

D. Adapter Tuning

Adapter tuning [48], [49], [50] is initially developed for fine-tuning pretrained language models for specific tasks, which involves integrating adapter layers that are exclusively trained during the fine-tuning phase, thereby preserving the original parameters of the comprehensive model. This method facilitates effective fine-tuning for targeted tasks while maintaining the extensive knowledge embedded within large pretrained models. The concept of adapter tuning has been extended to vision foundation models to support fine-tuning for visual tasks like classification [51] and segmentation [52]. To counter the increased inference overhead from adding network layers, low-rank adaptation methods [53], [54] employ a low-rank matrix. This approach significantly reduces the parameter number, adjusting the original weight matrix for more efficient fine-tuning.

Our methodology adopts adapter tuning combined with low-rank decomposition for GFS-Seg, preserving pretrained model parameters for base class knowledge extraction and optimizing the low-rank matrix for novel class learning. This approach not only achieves structural separation but also minimizes the overfitting problem common in full-parameter fine-tuning, making it particularly suited for GFS-Seg applications.

III. Approach

The overview of the proposed DFSA-Net framework is shown in Fig. 3. It is designed with a dual-phase training process to separate the feature learning for base and novel classes. Phase 1 focuses on training with base class annotations, employing the FSD to mitigate background feature interference by partitioning semantic tasks into foreground aggregation and multiclass semantic refinement. In Phase 2, the DLA is introduced to maintain the pretrained model's integrity and address overfitting through low-rank parameter adjustments for novel class adaptation. A key strategy during this fine-tuning phase involves applying SS to the base classes' foreground, thus improving the model's ability to learn novel classes. For inference, DFSA-Net follows the conventional semantic segmentation approach, taking an image as input and outputting a comprehensive segmentation mask for all classes by merging the outputs from both branches.

A. Problem Setup and Overview

The primary objective of GFS-Seg is to learn a model capable of segmenting images across both base and novel classes. In GFS-Seg, we have an ample collection of annotated images for N_b base classes, represented as $\mathbf{C^b} = \{c_1, \ldots, c_{N_b}\}$, and a limited set of K labeled support images (in the K-shot scenario) for each of the N_n novel classes $\mathbf{C^n} = \{c_{N_b+1}, \ldots, c_{N_b+N_n}\}$. The background category c_0 represents pixels not belonging to any target classes. The challenge in GFS-Seg involves distinguishing both base and novel classes from the background, targeting segmentation across N_b+N_n+1 classes.

To achieve this, DFSA-Net employs a two-phase training strategy as depicted in Fig. 3. Initially, in Phase 1, the model is trained with ample annotations for base classes to accomplish segmentation across $(N_b + 1)$ classes $(\mathbf{C^b} \cup c_0)$, utilizing a dilated ResNet-101 backbone encoder and the FSD. The proposed FSD separates semantic tasks into foreground estimation and multiclass refinement to suppress background feature interference. In Phase 2, the DLA is introduced, aiming to maintain the pretrained model's integrity and reduce overfitting through low-rank parameter adjustments for novel class adaptation. A notable DLA feature is SS on the base classes' foreground to improve learning for novel classes. Then, the model combines predictions from both base and novel class decoders for comprehensive pixel-wise prediction across all N_b+N_n+1 classes $(\mathbf{C^b}\cup\mathbf{C^n}\cup c_0)$, achieving targeted segmentation performance across all classes.

B. Foreground-Semantic Decoder

To mitigate background feature interference and bolster the learning for critical foreground category discrimination, we introduce the FSD. The FSD segments semantic tasks into foreground aggregation and multiclass semantic refinement, as illustrated at the top of Fig. 3. Specifically, for an image $I \in \mathbb{R}^{3\times H\times W}$ with height H and width W, the FSD processes

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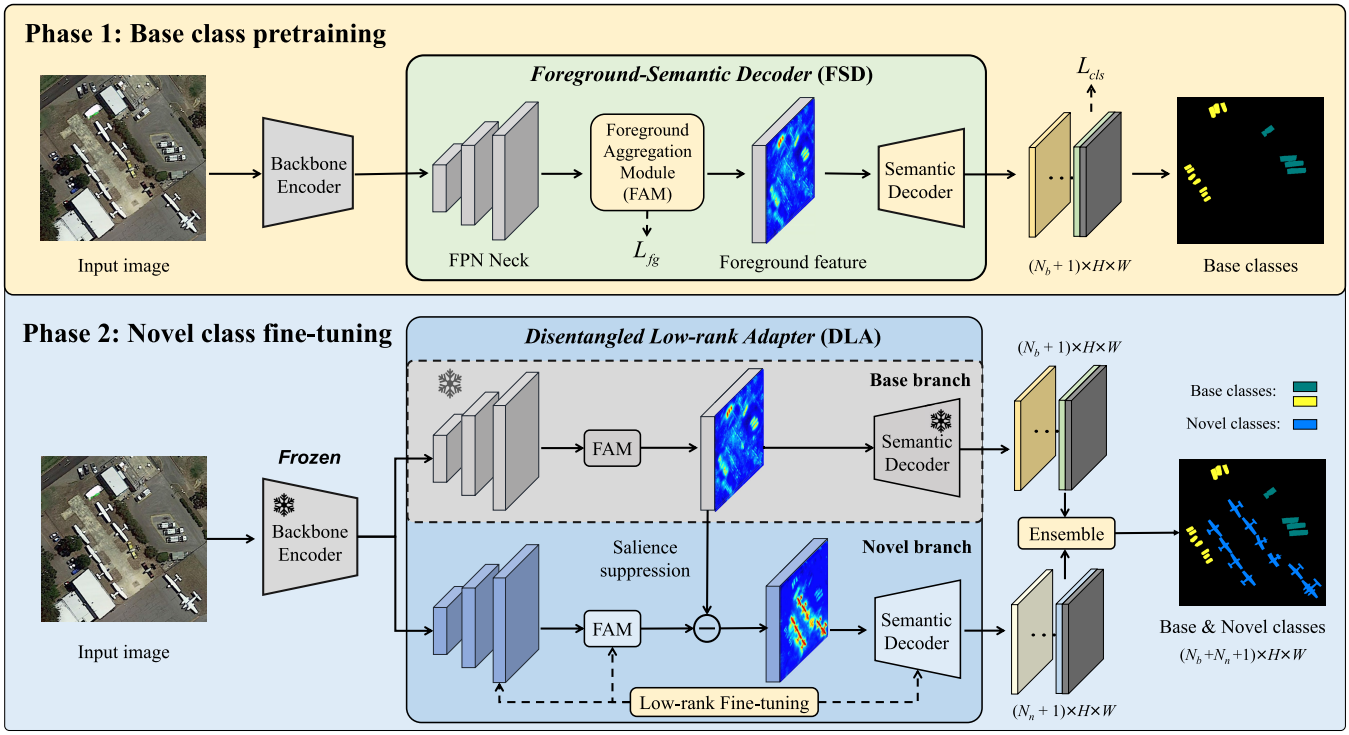


Fig. 3. Flowchart of the DFSA-Net, which consists of two training phase. In phase 1, our model is trained with sufficient base class annotations. To suppress background feature interference, we introduce the FSD by decomposing semantic tasks into foreground aggregation and multiclass semantic refinement. In Phase 2, the DLA is introduced to maintain the pretrained model's integrity and address overfitting through low-rank parameter adjustments for novel class adaptation. Notably, during the fine-tuning phase, we apply SS to the foreground of base classes to enhance the learning of novel classes.

multiscale feature maps $\{S^i \in \mathbb{R}^{2^{i-1}D_i\times(H/2^{i+1})\times(W/2^{i+1}}0\}_{i=1}^4$ provided by the backbone network, where D_i is the base channel number of stage i. Instead of directly adopting the backbone feature S^4 for decoding, we additionally leverage an FPN (see [13]) to enhance the fusion of high-level contextual information and low-level texture feature, which produces feature projection $\{F^i \in \mathbb{R}^{C\times(H/2^{i+1})\times(W/2^{i+1})}\}_{i=1}^4$. The FPN neck is mathematically described as

$$F^i = \begin{cases} \theta^i\left(S^i\right), & i = 4 \\ \theta^i\left(S^i\right) + \text{upsample}\left[\zeta^i\left(F^{i+1}\right)\right], & i = 1, 2, 3 \end{cases} \tag{1}$$

where C indicates the projected channel number, θ^i and ζ^i function as transformations for dimension reduction, executed through a sequence of 3×3 convolution, ReLU activation, and batch normalization, with the upsample operation referring to bilinear interpolation. After the FPN Neck, the model achieves high-resolution F^1 feature maps, integrating low-level texture and high-level semantic features.

Inspired by attention mechanism [36], [46], the FSD incorporates a foreground aggregation module (FAM) that enriches the features by aggregating foreground information, as depicted in Fig. 4. Specifically, FAM first transforms the FPN feature F^1 into F' through a 3×3 convolutions, and subsequently generates a foreground mask $\mathbf{M}_{\text{fg}}^{\text{base}} \in \mathbb{R}^{1\times H'\times W'}$, which can be formulated as

$$\mathbf{M}_{\text{fg}}^{\text{base}} = \delta\left(F'\right) \tag{2}$$

where $H' = H/4$, $W' = W/4$ denotes the height and width of feature map, δ represents a 1×1 convolution and bilinear

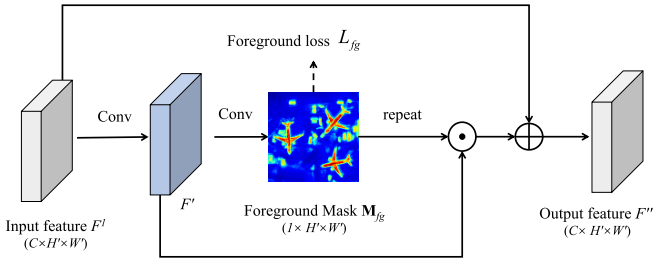


Fig. 4. Detail structure of the proposed FAM.

interpolation, followed by a sigmoid function. Notably, the foreground mask prediction is supervised by ground-truth mask, which is introduced in Section III-D. Then, we incorporate the foreground context into the newly generated feature map F'' through a residual attention structure as follows:

$$F'' = F^1 + \mathcal{R}\left(\mathbf{M}_{\text{fg}}^{\text{base}}\right) \odot F'. \tag{3}$$

Here, $\mathcal{R}(\cdot)$ refers to the repeat operation that expands the dimensionality from $\mathbb{R}^{1\times H'\times W'}$ to $\mathbb{R}^{C\times H'\times W'}$, and $\odot$ signifies the Hadamard product, i.e., element-wise multiplication. Subsequently, the FSD processes the foreground feature map F'' through a semantic decoder for multiclass refinement. The semantic decoder uses a series of 3×3 convolutions, bilinear interpolation, and then applies a softmax function, ultimately generating the segmentation map $\mathbf{M}_{\text{cls}}^{\text{base}} \in \mathbb{R}^{(N_{\text{base}}+1)\times H'\times W'}$ through bilinear interpolation. By segmenting the segmentation task into foreground aggregation and multiclass semantic refinement, FSD not only improves the separation between foreground objects and the background but also lightens the

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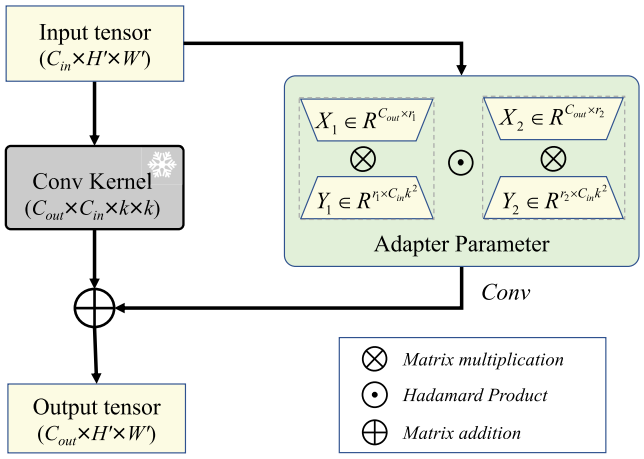


Fig. 5. Illustration of low-rank adaptation layer in the proposed DLA.

load on the semantic decoder. This enables it to focus more on interpreting critical foreground category discrimination.

C. Disentangled Low-Rank Adapter

In the novel class fine-tuning phase, as highlighted in Section IV, relying solely on transfer learning tends to lead to the degradation of base classes and overfitting of novel classes. To tackle these issues, we introduce the DLA specifically for fine-tuning novel classes, illustrated at the bottom of Fig. 3. To prevent base class degradation, we keep the parameters of the backbone and base pretrained branch frozen and add a trainable adapter branch for novel classes, ensuring stable outcomes for base classes throughout the fine-tuning stage. To alleviate the overfitting phenomenon on novel class segmentation, we design a disentangled fine-tuning strategy for a novel class inspired by adapter-tuning techniques prevalent in both vision and language models. The trainable adapter is added outside the frozen structure, thereby preserving the original landcover knowledge in the original model for transfer learning.

Specifically, within the novel branch, the FPN, FAM, and semantic decoder are restructured by substituting traditional convolution layers with our designed low-rank adaptation layers. The input and output of low-rank adaptation layer are the same as their original convolution layer. These layers approximate the learnable kernel weights $\widetilde{\boldsymbol{K}}$ using low-rank matrices. Fig. 5 shows the low-rank adaptation layer by reconstructing the learnable convolution kernel matrix through the Hadamard product, which can be formulated as follows:

$$\begin{aligned} \widetilde{\boldsymbol{K}} &= \text{reshape}(\boldsymbol{X_1 Y_1} \odot \boldsymbol{X_2 Y_2}) \\ F_{\text{output}} &= \text{Conv}(\widetilde{\boldsymbol{K}}, F_{\text{input}}) + \text{Conv}(\boldsymbol{K}, F_{\text{input}}) \end{aligned} \tag{4}$$

where $F_{\text{input}} \in \mathbb{R}^{C_{\text{in}}\times h\times w}$, $\boldsymbol{X_1} \in \mathbb{R}^{C_{\text{out}}\times r_1}$, $\boldsymbol{X_2} \in \mathbb{R}^{C_{\text{out}}\times r_2}$, $\boldsymbol{Y_1} \in \mathbb{R}^{r_1\times(C_{\text{in}}\times k\times k)}$, $\boldsymbol{Y_2} \in \mathbb{R}^{r_2\times(C_{\text{in}}\times k\times k)}$, and $\widetilde{\boldsymbol{K}} \in \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}\times k\times k}$. To fulfill the low-rank criteria, we set the ranks of our trained weights, r_1 and r_2, to be much less than the minimum of C_{in} and C_{out}, i.e., $r_1, r_2 \ll \min(C_{\text{in}}, C_{\text{out}})$. This approach significantly enhances the model's optimization efficiency and reduces memory consumption. Unlike traditional full-parameter fine-tuning, which is susceptible to overfitting in few-shot scenarios, our model mitigates this issue by training a smaller number of parameters. Furthermore, to preserve the model's original knowledge at the start, $\boldsymbol{X_2}$ is zero-initialized, while other low-rank matrices undergo normal initialization. The novel branch is then progressively updated to better accommodate segmentation of novel classes. In fact, network compression often utilizes similar low-rank decomposition techniques like Tucker and CP decomposition [55], [56] to find approximations within low-rank subspace. To simplify computations and aid in reconstructing higher dimensions, we use the Hadamard product strategy as referenced in (4).

To enhance the learning of novel classes, we apply SS to the foreground of base classes during the fine-tuning phase. In particular, the foreground mask $\mathbf{M}_{\text{fg}}^{\text{novel}} \in \mathbb{R}^{1\times H'\times W'}$, utilized for aggregation in the novel class branch, is derived from the difference between the prediction output and the base class foreground mask $\mathbf{M}_{\text{fg}}^{\text{base}}$ as follows:

$$\mathbf{M}_{\text{fg}}^{\text{novel}} = \delta_{\text{ada}}\left(F'_{\text{ada}}\right) - \mathbf{M}_{\text{fg}}^{\text{base}} \tag{5}$$

where δ_{ada} and F'_{ada} are obtained through the low-rank adaptation layer substitution of (2) in the novel branch. Importantly, since $\boldsymbol{X_2}$ is initialized to zero, the initial prediction output matches the base class foreground mask $\mathbf{M}_{\text{fg}}^{\text{base}}$, resulting in $\mathbf{M}_{\text{fg}}^{\text{novel}} = 0$ during the initial stage. This ensures that the foreground is exclusively composed of elements from novel classes. Such a focused foreground composition significantly bolsters the learning process for these novel classes, facilitating more effective and targeted training outcomes.

Similar to the base branch, the novel branch processes the foreground feature map F'' through a semantic decoder for multiclass refinement. The semantic decoder uses a series of 3×3 convolutions and then applies a softmax function, ultimately generating the segmentation map $\mathbf{M}_{\text{cls}}^{\text{novel}} \in \mathbb{R}^{(N_{\text{novel}}+1)\times H\times W}$ through bilinear interpolation. By segmenting the segmentation task into foreground aggregation and multiclass semantic refinement, FSD not only improves the separation between foreground objects and the background but also lightens the load on the semantic decoder. This enables it to focus more on interpreting critical foreground category discrimination.

D. Loss Function

In the first phase, the entire network is trained on sufficient base data to optimize the parameter in backbone and FSD. The training loss function contains two parts:

$$\mathcal{L}_{\text{stage1}} = \mathcal{L}_{\text{cls1}} + \lambda \mathcal{L}_{\text{fg1}} \tag{6}$$

where $\mathcal{L}_{\text{cls}}$ represents the pixel-wise multiclass cross-entropy loss, and $\mathcal{L}_{\text{fg}}$ represents the pixel-wise foreground binary cross-entropy loss, which are calculated as follows:

$$\mathcal{L}_{\text{cls1}} = \sum_{n=0}^{N_b} y_n \log\left(\mathbf{M}_{\text{cls},n}^{\text{base}}\right) \tag{7}$$

$$\mathcal{L}_{\text{fg1}} = (1-y_0)\log\left(\mathbf{M}_{\text{fg}}^{\text{base}}\right) + y_0 \log\left(1-\mathbf{M}_{\text{fg}}^{\text{base}}\right) \tag{8}$$

where $y_n = 1$ indicates that the pixel has the ground-truth class label n, otherwise $y_n = 0$. $\mathbf{M}_{\text{cls},n}^{\text{base}}$ denotes the probability

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that pixel belongs to class n according to the base model prediction, while $\mathbf{M}_{\text{fg}}^{\text{base}}$ indicates the probability of the pixel being identified as foreground.

In the second phase, we maintain all the modules trained in the first phase in a frozen state and only fine-tuning the low-rank convolution parameters for the novel and background classes. Similarly, the loss function is defined as

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{cls2}} + \lambda \mathcal{L}_{\text{fg2}} \tag{9}$$

where $\mathcal{L}_{\text{cls2}}$ and $\mathcal{L}_{\text{fg2}}$ is calculated as follows:

$$\mathcal{L}_{\text{cls2}} = y_0' \log\left(\mathbf{M}_{\text{cls},0}^{\text{novel}}\right) + \sum_{n=N_b+1}^{N_b+N_n} y_n \log\left(\mathbf{M}_{\text{cls},n}^{\text{novel}}\right) \tag{10}$$

$$\mathcal{L}_{\text{fg2}} = \left(1 - y_0'\right) \log\left(\mathbf{M}_{\text{fg}}^{\text{novel}}\right) + y_0' \log\left(1 - \mathbf{M}_{\text{fg}}^{\text{novel}}\right) \tag{11}$$

where $y_0' = 1$ indicates that the pixel belongs to the base class or background class. $\mathbf{M}_{\text{fg}}^{\text{novel}}$ signifies the probability of the pixel being classified as foreground within the novel classes. Compared to traditional full-class fine-tuning, our disentangled training loss for base and novel class branches enables the independent learning of features for each category. This approach effectively tackles the issues of base class degradation and novel class overfitting during the fine-tuning phase.

E. Inference Ensemble

After passing through two disentangled branches, the model produces a base classes' prediction score mask $\mathbf{M}_{\text{cls}}^{\text{base}} \in \mathbb{R}^{(N_b+1)\times H\times W}$ and a novel classes' prediction mask $\mathbf{M}_{\text{cls}}^{\text{novel}} \in \mathbb{R}^{(N_n+1)\times H\times W}$. The challenge lies in integrating these two sets of predictions. Initially, pixels belonging to both the novel classes $\mathbf{C^n}$ and the background class c_0 are categorized under the background. However, with the incorporation of novel classes in the subsequent phase, the semantic interpretation of the background changes as it no longer incorporates pixels from novel classes. This semantic shift phenomenon blurs the distinction between the novel classes and the background class, thus necessitating a careful merge of the results. To solve this problem, we propose an ensemble strategy as follows:

$$\mathbf{M}_{\text{cls}}^{\text{final}} = \begin{cases} c_0, & \text{if } \max\limits_{c_i \in \mathbf{C^b}} \mathbf{M}_{\text{cls},i}^{\text{base}} < \mathbf{M}_{\text{cls},0}^{\text{base}} \text{ and} \\ & \max\limits_{c_j \in \mathbf{C^n}} \mathbf{M}_{\text{cls},j}^{\text{novel}} < \mathbf{M}_{\text{cls},0}^{\text{novel}} \\ c_i, & \text{if } \max\limits_{c_i \in \mathbf{C^b}} \mathbf{M}_{\text{cls},0}^{\text{base}} \le \mathbf{M}_{\text{cls},i}^{\text{base}} \\ c_j, & \text{if } \max\limits_{c_i \in \mathbf{C^b}} \mathbf{M}_{\text{cls},i}^{\text{base}} < \mathbf{M}_{\text{cls},0}^{\text{base}} \text{ and} \\ & \max\limits_{c_j \in \mathbf{C^n}} \mathbf{M}_{\text{cls},j}^{\text{novel}} \ge \mathbf{M}_{\text{cls},0}^{\text{novel}}. \end{cases} \tag{12}$$

Here, $\mathbf{M}_{\text{cls}}^{\text{final}} \in \mathbb{R}^{1\times H\times W}$ is the final multiclass segmentation mask. $\mathbf{M}_{\text{cls},i}^{\text{base}}$ represents the prediction score mask for the ith class from the base decoder, while $\mathbf{M}_{\text{cls},j}^{\text{novel}}$ indicates the prediction score mask for the jth class from the novel decoder for a given pixel. $\mathbf{M}_{\text{cls},0}^{\text{base}}$ and $\mathbf{M}_{\text{cls},0}^{\text{novel}}$ correspond to the predicted probabilities of the background class by the base and novel decoders, respectively. Intuitively, (12) classifies a pixel as background if both decoders identify the pixel as such. Furthermore, it assigns to the foreground prediction of the base classifier, which benefits from extensive annotation. Furthermore, if a pixel is identified as background by the base decoder and as foreground by the novel decoder, it is subsequently classified to the novel class.

TABLE I
NOVEL CLASS SETTING ON DIFFERENT DATASETS

Dataset	Novel classes
NWPU VHR-10	airplane, baseball diamond, tennis court
iSAID splits 1	ship, storage tank, baseball diamond, tennis court, basketball court
iSAID splits 2	ground field track, bridge, large vehicle, small vehicle, helicopter
iSAID splits 3	swimming pool, roundabout, soccer ball field, plane, harbor

IV. EXPERIMENTS

In this section, we first introduce two datasets and evaluation metrics for assessing our method. Next, the implementation details and the parameter analysis are provided. Finally, we conduct experiments to compare our DFSA-Net against other FS-Seg methods and deliver a comprehensive ablation study to analyze the impact of our components.

A. Datasets and Evaluation Metrics

Currently, there is a scarcity of publicly accessible datasets for FS-Seg in remote sensing imagery with several limitations as introduced in Section I. In our experiment, we utilize the NWPU VHR-10 and iSAID [57] datasets, which are widely used in aerial image segmentation. Notably, we set the novel class segmentation masks to the background in the base training stage. During the fine-tuning phase, we randomly select K-shot training images for each novel class. Unlike previous approaches to FS-Seg [4], [6] that depended on cropped patches focusing on the foreground, we train our model using full images across all classes.

1) NWPU VHR-10: The NWPU VHR-10 dataset [58] includes 800 high-resolution satellite images from Google Earth and the Vaihingen dataset [59], featuring 650 positive images and 150 negative images with only background. This dataset provides annotation of object bounding boxes and instance masks, making it suitable for object detection and instance segmentation tasks. It categorizes landcover into ten classes: bridge (C1), harbor (C2), ground track field (C3), ship (C4), storage tank (C5), basketball court (C6), vehicle (C7), airplane (C8), baseball diamond (C9), and tennis court (C10). For our GFS-Seg endeavor, we classified three classes as novel—airplane, baseball diamond, and tennis court—with all other categories being designated as base classes, as outlined in Table I. In our experimental setup, we utilized 650 labeled positive images, dividing them into two-thirds (433) for training and one-third (217) for testing to ensure a uniform framework for comparison. The segmented dataset is available at https://github.com/qixiong-wang/mmfewshot-DFSA to stimulate additional studies and comparative analyses.

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2) iSAID: The iSAID [57] dataset is the largest aerial instance segmentation dataset, containing 2806 high-resolution remote sensing images. The dataset is divided into 1411 training images, 458 validation images, and 937 test images. For our study, we only used the training and validation sets for both training and evaluation purposes due to the unavailability of the ground truth for the test set. The images in iSAID come from a variety of sensors and platforms and vary in resolution, with sizes ranging from 800 × 800 to 4000 × 13 000 pixels. The iSAID dataset contains 15 categories, i.e., ship, storage tank, baseball diamond, tennis court, basketball court, ground field track, bridge, large vehicle, small vehicle, helicopter, swimming pool, roundabout, soccer ball field, plane, and harbor. In line with [4], we utilized a cross-validation method to test our model by designating five classes as test categories and the remaining ten as training categories. The specific distribution of classes for training and testing is detailed in Table I.

3) Evaluation Metrics: To quantitatively evaluate the performance of our model, we utilize the mean intersection over union (mIOU) metric in our experiments, aligning with methodologies adopted in previous studies [7], [27]. The mIOU metric provides a comprehensive measure of the model's accuracy by calculating the average ratio between the intersection and the union of the predicted and ground-truth segmentation masks across all classes

$$\text{mIOU} = \frac{1}{K}\sum_{k=1}^{K}\frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k + \text{FN}_k}. \tag{13}$$

Here, K represents the total class number in the dataset. True positives (TP) are the pixels accurately classified for a class, false positives (FP) are the pixels wrongly classified as part of a class, and false negatives (FN) are the pixels not recognized as part of their actual class.

B. Implementation Details

In our experiments, both the base pretraining and novel fine-tuning are conducted on the training set, with evaluations carried out on the test set. Importantly, during the base training stage, the pixels of novel classes are considered as background. In the novel class fine-tuning phase, we mimic the few-shot setting by randomly selecting K images for each novel class from original training set. For the NWPU VHR-10 dataset, we resize and pad input images to 896 × 896. To prevent overfitting, we use random flipping (with a 0.5 probability) as data augmentation. For the iSAID dataset with larger image scale, we crop the image into 896 × 896 with stride 512 following [43], [60] for training. During inference, we use sliding windows with size 896 × 896 and stride 512 to cover the entire image for seamless prediction, averaging multiple predictions to obtain the final result. To ensure stability of our results, we conduct each experiment five times using different random seeds and report the average values.

For implementation details, the framework is built on Pytorch1.11 and mmsegmentation [61]. All the experiments are conducted on a server with NVIDIA GeForce RTX 3090 (24 GB). We adopt stochastic gradient descent (SGD) algorithm as the optimizer. During the base pretraining on NWPU VHR-10 dataset, our model is optimized with a batch size of 8 on two GPUs. It is trained for 6k and 2k iterations with learning rates of 0.01 and 0.001, respectively. For iSAID dataset, it is trained for 80k and 40k iterations with learning rates of 0.02 and 0.002 on four GPUs. To slow the model's early overfitting of minibatch, we applied a warm-up strategy in the first 1500 iterations. In the novel fine-tuning phase, the parameters are initialized with base pretraining weights with additional fine-tuning for 20k iteration with a learning rate of 0.001 on one GPU. On the setting of hyperparameter, λ is set to 0.8 to balance foreground and multiclass segmentation, and the rank parameter in DLA convolution is set to $r_1 = r_2 = 64$.

C. Experiments on NWPU VHR-10 Dataset

1) Comparisons With State of the Art: To evaluate the performance of our DFSA-Net, we compare our method with six recently proposed approaches, which can be divided into FS-Seg-based and GFS-Seg-based. For FS-Seg-based methods, we adapted CANet [5], PFENet [30], and SDM [4] following [7], which generates a class prototype by averaging the features of pixels belonging to that class across all training images to segment the query image. For GFS-Seg-based methods, we reimplemented FT [32], CAPL [7], and POP [8] for aerial image segmentation. The comparative results are detailed in Table II. Additionally, we include the baseline pretraining performance, employing a straightforward ResNet101 alongside a basic FPN decoder. It can be observed that our DFSA-Net excels in adapting to the segmentation of new classes under 1-, 5-, and 10-shot settings, outperforming other comparative methods. Specifically, the mIOU on the novel class of DFSA-Net reaches 27.76% and 36.12% for 1- and 5-shot settings, respectively. This signifies a notable enhancement over its closest competitors by margins of 4.2% and 4.32%. Notably, FS-Seg models, such as CANet, PFENet, SDM, and HMRE, significantly underperform compared to GFS-Seg models. This demonstrates the limitation in the episodic training/testing strategy of FS-Seg, which aims to identify pixels of a single new class based on the prior knowledge from support samples. Hence, distinguishing novel classes under the interference of base classes in GFS-Seg settings becomes more challenging.

Relative to GFS-Seg models, the performances of FT [32] and CAPL [7] demonstrate the phenomenon of base class degradation in GFS-Seg tasks, with a notable decrease for base classes. In contrast, our DFSA-Net markedly surpasses the baseline FT model. This indicates that DFSA-Net successfully mitigates the problems of base class degradation and novel class overfitting. Moreover, compared to CAPL, which leverages contextual co-occurrence cues to enhance prototypes, DFSA-Net achieves improvements of 6.81%, 5.06%, and 6.54% in mIOU for base, novel, and all classes, respectively, under the 1-shot setting. This demonstrates the advantages of our disentangled structure for base and novel class learning. Compared with the recently proposed POP [8], which mitigates base class degradation by encouraging the orthogonality between the learned prototypes, our method still achieves 1.88% and 1.94% improvement in mIOU for total class

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TABLE II
EXPERIMENT RESULTS ON THE NWPU VHR-10 DATASET

	Method	Base class								Novel Class				Total
		C1	C2	C3	C4	C5	C6	C7	mean	C8	C9	C10	mean	
	Pretrained Baseline	57.36	68.28	89.25	44.21	59.51	73.81	46.35	62.68	-	-	-	-	-
1-shot	CANet [5]	2.37	4.24	8.23	6.72	3.32	5.47	5.16	5.07	1.53	5.76	4.33	3.87	4.71
	PFENet [30]	7.43	8.04	13.36	12.67	11.08	9.72	12.40	10.67	4.24	9.45	8.27	7.32	9.66
	SDM [4]	8.37	9.20	16.37	13.26	9.55	12.03	11.42	11.45	4.38	11.82	9.56	8.58	10.59
	HMRE [31]	7.52	9.58	12.87	11.36	10.29	12.64	10.74	10.71	5.21	10.27	9.28	8.25	9.97
	FT [32]	43.60	65.88	85.68	26.18	54.05	71.52	43.86	55.82	12.03	39.74	15.98	22.59	45.85
	CAPL [7]	46.54	62.73	84.25	28.57	55.93	72.23	44.70	56.42	12.37	39.76	15.98	22.70	46.05
	POP [8]	57.13	**68.02**	88.63	43.50	**59.23**	73.57	46.34	62.35	13.73	38.54	18.42	23.56	50.71
	DFSA-Net (Ours)	**59.61**	67.53	**89.88**	**44.72**	58.64	**74.84**	**47.40**	**63.23**	**16.58**	**43.70**	**23.02**	**27.76**	**52.59**
5-shot	CANet [5]	2.84	4.90	8.85	6.83	3.89	6.11	5.83	5.61	1.75	4.28	5.19	3.74	5.05
	PFENet [30]	7.72	8.67	14.55	11.52	13.15	10.27	13.23	11.30	5.21	11.82	9.25	8.76	10.54
	SDM [4]	8.35	9.86	13.59	14.50	9.84	14.12	12.98	11.89	5.38	12.91	9.64	9.31	11.12
	HMRE [31]	8.14	10.25	14.32	13.68	10.75	12.48	12.54	11.73	5.52	12.37	9.46	9.11	10.95
	FT [32]	42.86	64.42	86.32	24.71	53.35	71.83	44.58	55.44	16.62	48.31	27.16	30.65	48.02
	CAPL [7]	48.42	63.03	87.42	32.29	56.31	72.84	45.25	57.94	15.72	47.83	27.43	30.32	49.65
	POP [8]	56.47	67.64	89.35	43.25	**59.67**	73.35	46.52	62.32	17.05	48.52	29.84	31.80	53.17
	DFSA-Net (Ours)	**58.37**	**68.04**	**90.11**	**44.69**	59.50	**73.82**	**48.16**	**63.24**	**20.35**	**53.18**	**34.83**	**36.12**	**55.11**
10-shot	CANet [5]	4.47	5.11	9.28	7.90	5.81	7.72	7.78	6.87	1.99	4.92	5.36	4.09	6.03
	PFENet [30]	8.46	8.75	14.98	12.56	14.21	10.70	14.07	11.96	5.34	12.36	9.73	9.13	11.12
	SDM [4]	9.14	10.26	15.53	14.75	11.04	14.29	14.12	12.73	5.58	13.42	10.41	9.80	11.85
	HMRE [31]	9.06	11.23	14.58	12.69	10.91	13.32	13.97	12.25	5.73	13.12	9.74	9.53	11.44
	FT [32]	43.35	64.73	86.63	28.62	54.52	71.61	44.67	56.30	21.35	52.84	31.16	35.12	49.95
	CAPL [7]	49.21	63.67	87.02	31.71	57.34	72.47	45.61	58.15	20.18	51.85	30.58	34.20	50.96
	POP [8]	56.67	67.64	89.65	43.75	58.67	73.35	46.83	62.37	23.51	54.78	32.54	36.94	54.73
	DFSA-Net (Ours)	**58.51**	**68.25**	**90.25**	**44.29**	**59.31**	**75.82**	**48.41**	**63.55**	**27.43**	**59.32**	**38.76**	**41.84**	**57.03**

under 1- and 5-shot settings, largely attributable to gains in new class performance. This highlights the strength of our proposed strategy in enhancing the focus on critical foreground categories.

2) Ablation Study of Module Combination: We conduct ablation studies to investigate the impact and efficiency of various components within the proposed network, and the efficiency is measured by the time cost and FLOPs for an input image with a size of $3 \times 896 \times 896$. Initially, we set vanilla FT [32] model as our baseline, which directly fine-tunes the network using support set. The results are depicted in Table III. Specifically, FAM indicates the incorporation of the FAM in our proposed FSD. DLA refers to our disentangled low-rank adapter, i.e., the disentangled branches for base and novel class learning. SS represents salience suppression operation within the DLA. It can be observed that after adopting FAM in decoder, the performance increases by 1.74% and 2.18% in mIOU on base and novel classes, respectively, with 1.08 ms additional time overhead. This shows the importance of filter the interference of background and focus on essential foreground categories' discrimination. Moreover, by introducing DLA for disentangled learning for base and novel class object, the model achieves 7.27% and 1.52% increment in mIOU on base and novel classes, respectively. This shows that DLA can well preserving the integrity of the pretrained model while preventing the overfitting on novel classes. Additionally,

TABLE III
ABLATION OF MODULE COMBINATION ON NWPU VHR-10 DATASET

Module			mIOU(%)			time(ms)	Flops(G)
FAM	DLA	SS	Base	Novel	Total		
×	×	×	55.41	30.65	48.02	26.10	242.49
✓	×	×	57.15	32.83	49.61	27.18	245.84
×	✓	×	62.68	32.17	53.34	27.33	253.45
✓	✓	×	**63.35**	34.56	53.85	28.44	265.31
✓	✓	✓	63.24	**36.12**	**55.11**	28.53	256.73

by integrating FAM and DLA together, our method significantly outperforms the baseline by 7.94% and 3.91% in mIOU on base and novel classes, respectively. Further, after adopting SS operation to enhance the learning on novel classes, the mIOU on novel classes further boosts to 36.12% with an additional 2.43 ms inference time. These experiments reveal that our modules not only complement each other but also can be integrated with an acceptable time cost.

3) Ablation Study of Ensemble Strategy: As shown in Fig. 3, our proposed DLA produces base prediction map $\mathbf{M}_{\text{cls}}^{\text{base}} \in \mathbb{R}^{(N_{\text{base}}+1)\times H\times W}$ and novel prediction map $\mathbf{M}_{\text{cls}}^{\text{novel}} \in \mathbb{R}^{(N_{\text{novel}}+1)\times H\times W}$. As illustrated in Section III-E, the background predictions from the base classifier may include novel class

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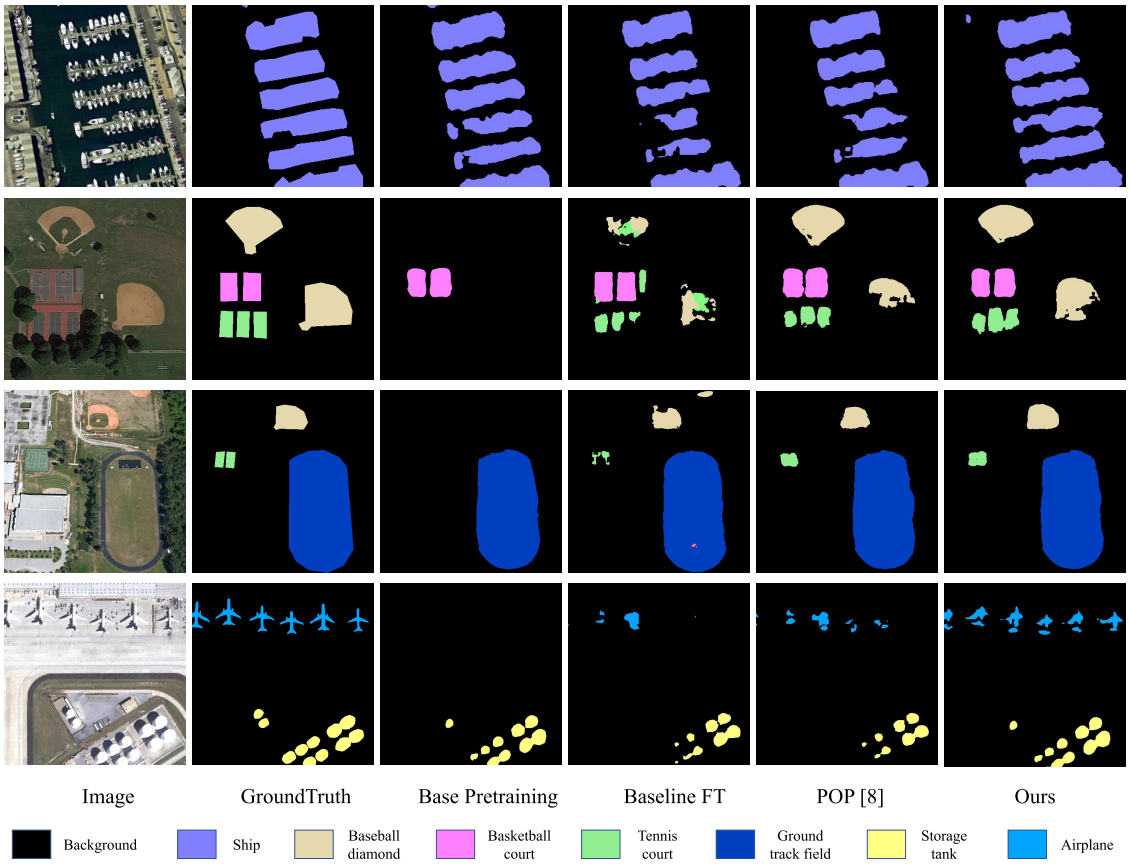


Fig. 6. Qualitative visualizations of segmentation results on image patches of the NWPU VHR-10 dataset under five-shot setting.

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT ENSEMBLE STRATEGY ON THE NWPU VHR-10 DATASET

Ensemble Method	Base	Novel	Total
Concatenate	61.65	**36.32**	54.01
BG Addition	61.87	35.63	53.79
Ours	**63.24**	36.12	**55.11**

TABLE V
ABLATION STUDY OF RANK PARAMETER r

Rank r	16	32	64	128	Full
Base	63.22	**63.33**	63.24	63.17	63.26
Novel	34.53	35.62	**36.12**	35.87	35.34
Total	54.59	55.07	**55.11**	55.04	54.86

pixels, and vice versa for the novel classifier. To address this problem, we tested three ensemble strategies for merging the predictions as shown in Table IV. Specifically, *Concatenate* simply combines the base and novel class prediction results to identify the highest confidence. *BG addition* averages the background prediction mask and find the most confident category. *Ours* represents the merge strategy illustrated in (12). It can be seen that our strategy achieves the highest total mIOU 55.11%. This attributes to our strategy's emphasis on maintaining accurate base class prediction, as the base classifier is trained on sufficient data.

4) Ablation Study of Hyperparameters: As mentioned in Section III-C, we substitute traditional convolution layers in novel branch with our designed low-rank adaptation layer to inherit the knowledge in base decoder, as well as prevent overfitting. In order to analyze the impact of rank r on segmentation accuracy, we conduct experiment to find the optimal hyperparameter rank r. The results are shown in Table V, "*full*" represents using traditional convolution layer the same as base branch. It can be seen that r merely has no impact on base class performance and the performance on novel class reaches the peak at 36.12% when $r = 64$. However, the performance drops when r is greater than 64 or full parameter. We conjecture that fine-tuning too many parameter under few-shot setting might cause overfitting to the training dataset.

As mentioned in Section III-D, the proposed DFSA-Net incorporates two types of loss at training stage, i.e., pixel-wise multiclass cross-entropy loss $\mathcal{L}_{\text{cls}}$ and foreground binary cross-entropy loss $\mathcal{L}_{\text{fg}}$, which is moderated by hyperparameter λ. Here, we conduct additional experiments to analyze the impact of weight parameters λ on the segmentation results as shown in Table VI. We observed a decrease of 1.77% in total mIOU when $\lambda = 0$, highlighting the significance of our proposed FAM. Generally, the model shows stable performance for different λ, with optimal results obtained at $\lambda = 0.8$. Based on the above observation, we choose the optimal settings throughout the experiments.

5) Qualitative Results: We also present a visualization of the segmentation results in Fig. 6. In the top row, unlike

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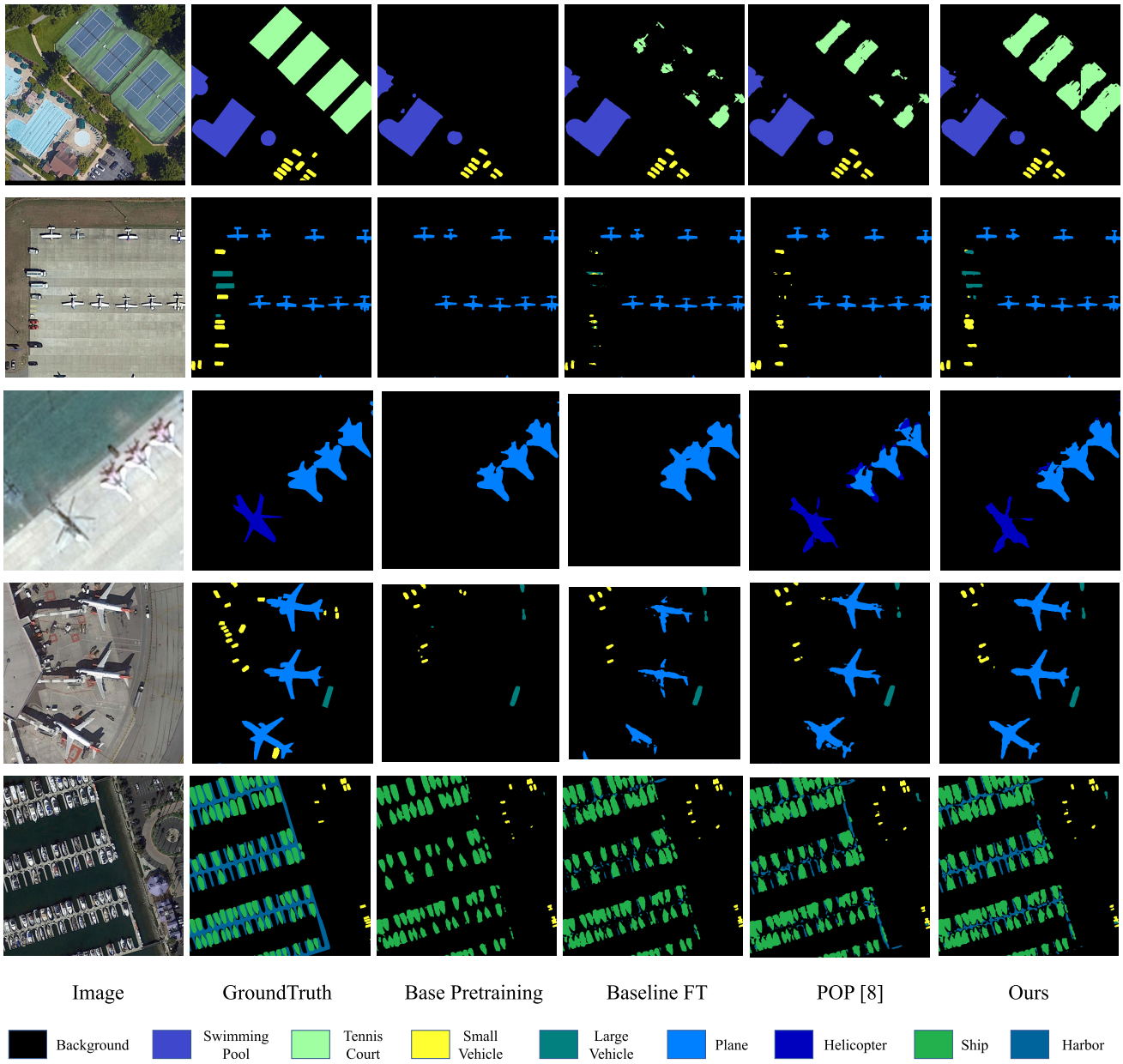


Fig. 7. Qualitative visualizations of segmentation results on image patches of the iSAID data set. (Top to bottom) examples are sampled from the validation set in split 1, 2, 3, respectively.

TABLE VI
ABLATION STUDY OF HYPERPARAMETER λ

λ	0	0.6	0.7	0.8	0.9	1
Base	62.68	62.96	63.12	**63.24**	63.17	63.05
Novel	32.17	34.67	35.86	**36.12**	35.62	35.33
Total	53.34	54.58	54.93	**55.11**	54.86	54.76

the baseline FT and POP [8], which show degradation in segmenting ships, our approach effectively maintains base class segmentation accuracy. This is achieved through our disentangled branch designed for separate base and novel class learning. In the middle row, while both the baseline model and POP struggle to differentiate between baseball diamonds and tennis courts, our method accurately identifies these novel classes. In the third row, the baseline model fails to precisely outline the tennis court, whereas our model achieves superior segmentation performance. For the bottom row, our model achieves better segmentation results on the novel airplane class, affirming our model's improved proficiency in recognizing novel foreground objects. These examples underscore the effectiveness of DLA and FAM in mitigating base class degradation and novel class overfitting.

D. Experiments on iSAID Dataset

For the iSAID dataset, we have selected six representative methods as comparative references, which comprise three FS-Seg methods and three GFS-Seg methods. The numerical results on the iSAID dataset are presented in Table VII, where the dataset is divided into three split for cross-validation. It can be seen that our model achieves the best results in

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TABLE VII
EXPERIMENT RESULTS ON THE iSAID DATASET

Split	Method	Split 1			Split 2			Split 3		
		Base	Novel	Total	Base	Novel	Total	Base	Novel	Total
	Pretrained Baseline	56.83	-	-	67.42	-	-	58.40	-	-
1-shot	CANet [5]	1.67	0.63	1.31	1.78	0.82	1.44	2.35	0.73	1.82
	PFENet [30]	3.38	0.87	2.50	2.92	1.04	2.31	2.61	1.17	2.15
	SDM [4]	3.43	1.31	2.71	2.73	1.17	2.21	2.89	1.35	2.36
	HMRE [31]	3.24	1.25	2.57	3.16	1.23	2.52	2.82	1.41	2.34
	FT [32]	48.92	15.19	37.72	59.73	12.14	43.86	52.73	12.85	39.42
	CAPL [7]	50.15	15.82	38.72	60.81	12.43	44.67	53.16	13.42	39.94
	POP [8]	56.39	16.25	42.98	67.28	12.85	49.13	58.13	13.61	43,31
	DFSA-Net (Ours)	**58.54**	**17.45**	**44.85**	**69.21**	**13.52**	**50.72**	**60.52**	**14.83**	**45.29**
5-shot	CANet [5]	2.32	0.84	1.82	2.15	1.21	1.84	2.84	1.08	2.26
	PFENet [30]	3.63	1.15	2.81	3.27	1.43	2.66	3.46	1.31	2.73
	SDM [4]	3.85	1.42	3.04	3.41	1.37	2.73	3.63	1.52	2.92
	HMRE [31]	4.12	1.45	3.24	3.57	1.46	2.86	3.51	1.48	2.82
	FT [32]	48.57	16.87	38.02	60.32	12.94	44.53	53.05	14.52	40.21
	CAPL [7]	49.82	17.64	39.14	61.54	13.32	45.41	53.40	14.77	40.21
	POP [8]	56.42	18.34	43.73	67.17	13.78	49.39	58.32	15.28	43.97
	DFSA-Net (Ours)	**58.57**	**20.68**	**45.94**	**69.13**	**14.91**	**51.04**	**60.66**	**16.62**	**45.94**
10-shot	CANet [5]	2.46	1.05	1.99	2.36	1.41	2.04	2.92	1.24	2.36
	PFENet [30]	3.67	1.24	2.86	3.52	1.62	2.89	3.62	1.42	2.89
	SDM [4]	3.92	1.47	3.10	3.48	1.53	2.83	3.92	1.62	3.15
	HMRE [31]	4.24	1.56	3.35	3.73	1.62	3.03	3.73	1.59	3.02
	FT [32]	50.32	20.46	40.37	61.52	14.56	45.87	54.21	17.57	42.03
	CAPL [7]	50.63	21.52	40.92	62.46	15.06	46.66	54.26	18.03	42.18
	POP [8]	56.58	23.58	45.58	67.23	16.87	50.44	58.56	19.36	45.49
	DFSA-Net (Ours)	**58.62**	**26.25**	**47.83**	**69.18**	**18.85**	**52.40**	**60.62**	**21.63**	**47.62**

various experimental conditions regardless of base or novel classes. Notably, the novel class performance metrics are lower than iSAID-5^i in SDM [4] for FS-Seg models, which can be attributed to different experimental setups. In our setup, models are tasked with identifying all classes within a test image, encompassing both base and novel classes. In contrast, the models are only required to detect pixels corresponding to a single novel class in iSAID-5^i, guided by support samples offering prior knowledge about the target class. Consequently, discerning novel classes in the context of GFS-Seg is more challenging due to the additional complexity of simultaneously recognizing multiple class types. It can be seen that the FS-seg methods merely cannot recognize specific landcover categories, achieving less than 4% in both base and novel classes. This demonstrates the gap between FS-Seg methods and general semantic segmentation.

Compared with GFS-seg methods, it can be seen that FT [32] suffers from severe base class degradation compared with pretrained baseline, with 7.91%, 7.69%, 5.67% decrement under 1-shot setting for split 1–3, respectively. In contrast, after equipped with the proposed FAM and DLA, our DFSA-Net achieves 58.54%, 69.21%, and 60.52% in base class mIOU, which indicates that DFSA-Net preserves the pretrained model's performance while effectively transferring knowledge to novel classes. Compared with competitive POP [8], DFSA-Net shows improvements of 1.2%, 0.67%, and 1.22% in novel class mIOU for the same splits under the 1-shot setting, which further increase to 2.34%, 1.13%, and 1.34% in the 5-shot setting. These results validate the effectiveness of our FAM and DLA approach, enhancing the model's adaptability and addressing overfitting challenges through precise low-rank adaptations for novel classes.

To achieve an intuitive comparison, samples of segmentation results of FT baseline, POP [8], and the proposed DFSA-Net are depicted in Fig. 7. The first row demonstrates that the baseline fine-tune and POP [8] fail to recognize part of base swimming pool and novel tennis court, whereas the proposed DFSA-Net makes relatively accurate predictions. Additionally, the second row presents DFSA-Net outperforms the baseline fine-tuning in scenes where small vehicles are intermixed with large vehicles, demonstrating the model's efficacy in foreground category discrimination in complex environments. This highlights the efficacy of our DLA, designed to learn base and novel classes separately, demonstrating its significant contribution to improving segmentation accuracy for both base and novel classes. In the third row, POP suffers from inter-class homogenization, which leads to a degradation in segmenting the base class airplane. In the last row, the examples of densely parked ships and harbors further affirm the model's improved proficiency in recognizing novel foreground objects, attributed to the strategic use of SS, thereby underscoring DFSA-Net's advanced segmentation capabilities in handling diverse and challenging scenarios.

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V. Conclusion

This study presents the DFSA-Net aimed at enhancing FS-Seg in aerial imagery, effectively bridging the gap between aerial image FS-Seg and standard semantic segmentation practices. In this framework, an FSD is proposed to reduce background interference by decomposing semantic segmentation into foreground aggregation and multiclass refinement. Additionally, the incorporation of a DLA alleviates the degradation of base classes and the overfitting to novel classes during the fine-tuning stage. Furthermore, our ensemble strategy combines the predictions from base and novel class decoders, significantly improving segmentation precision. Experimental evaluations on the NWPU and iSAID datasets have validated the effectiveness of DFSA-Net, showcasing its superiority over existing few-shot and GFS-Seg methods.

For future work, we will explore efficient prompt fine-tuning with a few annotated samples using visual foundation models, such as the segment anything model [62], [63]. Leveraging the knowledge of large foundation models for FS-Seg in aerial imagery is a promising direction.

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